FOOD

BEYOND THE CLASSROOM PLATE

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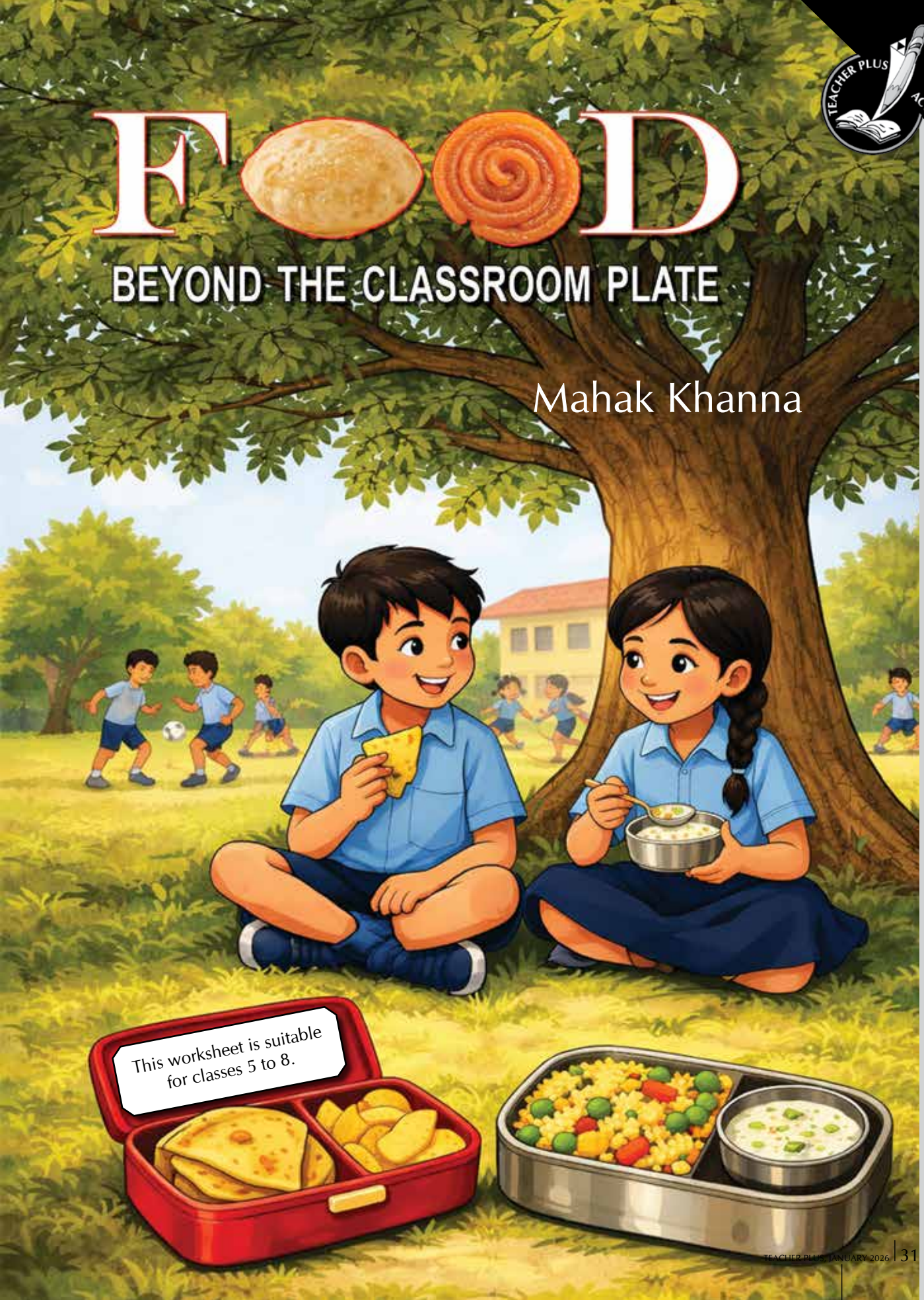

This worksheet is suitable for classes 5 to 8.

The case of the two lunchboxes

Read the detailed case carefully and analyze the situation
Every day at lunchtime, Arav and Kaira sit together under the neem tree in their school courtyard. Arav opens his shiny red tiffin box, which almost always contains *two ghee parathas* folded neatly, and a *small packet of potato chips*. He loves the crunch and says it helps him "stay awake" in boring classes.

Kaira's lunchbox is different each day: sometimes it's *vegetable pulao with peas and carrots*, sometimes *rotis with paneer sabzi*, and often a small *steel bowl of curd*. She says her mother tells her, "Colourful foods keep your brain sharp."

Over the last month, their class teacher has noticed a pattern:
- In the last two periods, Arav often slouches, yawns repeatedly, and sometimes struggles to focus on group work.
- Kaira, on the other hand, participates actively, asks questions, and even helps others complete their tasks.

One Wednesday, after a long science period, Arav complained of a "heavy stomach" and asked to visit the medical room. The nurse, after checking him, simply asked, "Beta, what did you eat today?"

Arav wondered if his favourite lunch could really be connected to how he felt later?

Meanwhile, Kaira remembered a science video they watched about how different nutrients break down at different speeds. She whispered to Arav, "Maybe your lunch gives quick energy but not *lasting* energy."

Their teacher, overhearing the conversation, smiled and said,
"Sometimes, food doesn't just fill the stomach… it controls how your mind works, how your body feels, and how long your energy lasts."

Based on the events described, what **connection** can you infer between the type of food eaten and afternoon alertness?

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Compare both lunchboxes and identify **which nutrient groups** Arav is lacking in his tiffin. Explain the reason for your answer.

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Suggest one specific and realistic change Arav can make to his lunch that will improve his afternoon concentration. Explain the scientific reason behind your suggestion.

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If *you* were the teacher investigating this pattern, what **evidence-based questions** would you ask Arav to confirm whether food is truly the cause of his behaviour (and not something else like sleep, stress, or illness)?

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Food classification – but with a twist

Take a look at the four food items below. All of them are commonly eaten by children your age:

Your task is NOT to label them as "healthy" or "unhealthy", but to analyze how each food behaves inside the body after you eat it.

Create two groups based on how long the energy from these foods is likely to last in the body:

- Group 1 – Quick energy, fades fast
- Group 2 – Slow, steady, long-lasting energy

Place the items accordingly.

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Justify your grouping by explaining how the body processes each item
(think: sugar breakdown, fiber content, protein, fats, refining, digestion time).

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Now imagine Arav (from Question 1) replaces *one* item in his lunchbox with something from Group 2. Predict one specific change that might occur in his behaviour during the afternoon classes – explain the *scientific reasoning* behind your prediction.

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Finally, choose one item from the list that students believe gives long-lasting energy but actually doesn't.

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The mystery of the missing nutrient

According to a *Times of India* feature, a large number of children in India suffer from **micronutrient deficiencies** – especially **iron, vitamin D, calcium, vitamin B_{12}, zinc, iodine**, and **omega-3**. Doctors say this is linked to modern habits like *skipping breakfast, rushed morning meals, eating leftover or packaged food, indoor lifestyles with little sunlight,* and *high snack consumption*. Children with such deficiencies often show signs like **fatigue, repeated infections, pale skin, low concentration, bone pain, muscle cramps, and slow growth**.
(Source: *Times of India – Common nutritional deficiencies seen in kids*)

Pick **two** of the deficiencies mentioned. For each, explain how it could affect a student's **school performance**.

A student eats: *refined bread sandwich, biscuits, chips, soft drink, and instant noodles*. Using the information from the article, identify **three deficiencies they are likely to face**, and explain **why**.

Create a **one-day meal plan** (breakfast–lunch–snack–dinner) that prevents at least **three** of the deficiencies.
For each meal, name the foods AND which nutrient they provide.

Interpreting a food map

Every food item travels through many steps before it reaches your plate. Below is a simplified food journey:
Farm → Transport → Wholesale Market → Storage/Cold Chain → Local Vendor → Home Kitchen → School Lunchbox

Recently, a news report highlighted major losses in fruits and vegetables in India because of breakdowns in cold storage facilities and lack of temperature-controlled transport.

Respond:
If the *storage/cold chain* suddenly fails in a city during summer, what type of foods (fruits/vegetables/milk products/others) will be affected the most? Explain **why** using scientific reasoning about spoilage and temperature.

Predict **two changes** you may observe in your local market (prices, freshness, availability).

Suggest **one change** you would make to the food journey to reduce losses when storage fails.

Explain how the failure of this system can indirectly affect a student's **nutrition** and **what might end up in their lunchbox**.

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Food chain thinking
You have a simplified food chain in a grassland:
Grass → Grasshopper → Frog → Snake → Hawk

Now imagine that due to changes in farming practices, farmers heavily spray pesticides to protect crops. These pesticides wash into nearby fields and are absorbed by the grass.

Answer:
(a) Explain which animal in the chain will end up with the highest concentration of the pesticide and why. Use the idea of *bioaccumulation*.

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(b) Predict **two changes** that may occur in the grassland ecosystem if frog populations drastically decline due to pesticide exposure.

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(c) Suggest **one environmentally safer farming practice** that could help protect this food chain.

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(d) Explain how this scenario connects back to humans – how can pesticide accumulation in animals ultimately affect *our food* and *our health*?

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Analyzing food labels
A sports drink and a flavoured milk carton are popular among students. Below are their nutrient values per 250ml:

Nutrient	Sports Drink	Flavoured Milk
Energy	160 kcal	120 kcal
Sugar	28 g	15 g
Protein	0 g	6 g
Calcium	0 mg	220 mg
Sodium	180 mg	80 mg

If a student wants **steady energy for studying after school**, which drink is more suitable? Give **two reasons** based on the table.

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Which drink will cause a **faster spike and crash in energy**? Use the sugar content to justify.

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If you had to improve one of these drinks to make it healthier, what **two changes** would you make and why?

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Explain **why reading only 'Energy (kcal)' is not enough** to judge whether a drink is good for you.

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Write a short response (6-8 lines):
If every apartment building grew vegetables on rooftops and balconies, what long-term changes could it bring to food quality, cost, and the environment?

Include at least:
- one benefit for **health**,
- one benefit for **cities/environment**, and
- one challenge society may face.

Write down your response in the space provided below.

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